

# Passing Coarse Marginal Checks Can Be Cheap: Persona Mixtures and Imprecise Treatment-Response Estimates in an LLM Persona Panel

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Review Revision v11 - Post-v10 Addendum - Not for Citation

**Review artifact identity.** Repository `yoheinakajima/synthetic-players`; branch `paper/explore-v11-reporting`; source commit `e61555a1094d5e563e4310bb3eb9a7b00f3eadd`; source docs/`paper/paper-draft.md`; base scientific freeze `paper-text-freeze-v10`. The PDF checksum is committed beside the generated file.

**STATUS: REVIEW REVISION v11 — POST-v10 ADDENDUM, NOT FOR CITATION.** The v10 scientific text freeze remains archived and unchanged. This v11 addendum incorporates the subsequent Explore Science reporting-completeness review and one new zero-call fixed-panel latent-propensity sensitivity; all sealed experimental artifacts and historical mechanical verdicts remain unchanged. Every sealed registration, adjudication, report, and precommitted discussion artifact remains unchanged. §5.2 quotes an exact excerpt of discussion text sealed and externally timestamped before the final experiment’s data existed (sha `1f1d7de9...e356`); the full sealed text remains in the supplement and is not edited. Draft history, review records, generated figures, and the correction ledger are public in the repository.

**Author:** Yohei Nakajima (Untapped Capital). Experiments executed by an autonomous pipeline (Replit Agent + ActiveGraph event-sourced engine). Attribution and reviewer-role disclosure: §8.

**Artifacts (public):** `github.com/yoheinakajima/synthetic-players` — anonymous clone + one-command zero-credential verifier; 4,576/4,576 Phase 4–5 runs replay byte-exact; registries hashed and externally anchored (GitHub timestamps + OpenTimestamps/Bitcoin).

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## Abstract

Large language models are increasingly used as synthetic research participants, but they are often validated by whether their marginal responses resemble published human data. We report a five-phase research program. Confirmatory claims from Phases 3–5 were registered before the data that adjudicated them and were mechanically evaluated from an event-sourced record; Phases 1–2 document post hoc instrument development and corrective re-adjudication rather than prospective confirmation. A fixed panel of sixteen lightweight persona-conditioned GPT-4.1 configurations passed preregistered broad-reference checks for condition-level cooperation in three of four repeated-game cells. Finite-opportunity-corrected point estimates place between-prompt standard deviations at 0.418–0.478 and 85%–96% of observed episode-level variation between prompt configurations. A fixed-panel Dirichlet–Jeffreys latent-propensity sensitivity, which propagates uncertainty from six-episode boundary cells, yields posterior median between shares of 63%–71% with 95% intervals spanning 49%–81%. The observed aggregate continuation-probability contrasts are +0.083 and +0.078 across two wording families; conservative exact simultaneous 95% intervals are  $[-0.171, +0.330]$  and  $[-0.181, +0.330]$  (rounded to three decimals). The treatment jointly changed the continuation process and the language communicating it, so incentive and framing channels remain undecomposed. Cell-level boundary classification is also method-sensitive: the historical seat-level rule classified 14/96 cells interior, a conservative exact episode-level interval classified 11/96, and a Dirichlet–Jeffreys sensitivity classified 19/96. Separate representation

experiments showed that, in the bare non-persona-conditioned configuration, replacing and repositioning one continuation sentence shifted cooperation from 0/40 to 37/40 on held-out decisions, and that semantic action labels made the bare configuration choose an action dominated by the payoffs in a registered conflict condition. Together, the findings identify a concrete validation failure mode: broad-reference marginal criteria can be satisfied through composition across concentrated prompt-conditioned policies without requiring the treatment-response object to be estimated. External review later exposed family-error, dependence, and boundary-uncertainty defects; post-adjudication sensitivities changed the scientific interpretation without rewriting the historical record. Human references are published and protocol-nonmatched, the results concern one fixed model-persona panel, and we do not claim human substitutability.

## 1. Introduction

Human behavioral experiments are slow and expensive; LLM calls are fast and nearly free. A growing literature reports that suitably conditioned LLMs produce data resembling human data—“algorithmic fidelity” [Argyle et al. 2023], “homo silicus” [Horton 2023], behavior “statistically indistinguishable from a random human” [Mei et al. 2024]—and a formal framework for evaluating *statistical realism* now exists [Xie et al. 2026]. Much of this evidence validates marginals: means, distributions, and aggregate replication.

Recent work shows that descriptive realism and causal fidelity can diverge at scale [Li & Ji 2026], that treating LLM outcomes as surrogates for human outcomes requires assumptions that marginal equivalence does not supply [Persson et al. 2026], and that intervention prompts can shift a model’s implied latent user even when explicit persona text is fixed [Lin et al. 2026]. Li and Ji additionally trace effect errors to intervention logic, outcome structure, and excessive attitude–behavior coupling, so the present paper does not claim mechanism-level explanation in general. Its narrower contribution is a specific fixed-panel composition mechanism in incentive-bearing strategic interaction, coupled to minimal representation interventions and an auditable correction record. We decompose the explicit persona panel into between-prompt dispersion, within-prompt variation, and prompt-indexed treatment response. The panel passes coarse marginal checks while producing small but imprecisely estimated continuation-probability contrasts, and most recorded variation lies between prompt configurations rather than within them. Sealed templates and matched procedures control explicit assignment, environment randomization, and execution. They do not establish latent-person invariance; Lin-style user drift and the observed composition pattern can coexist.

Here, “cheap” denotes evidentiary economy rather than only low API cost: the broad-reference marginal checks could be passed without estimating the treatment-response object they might be taken to validate.

**Contributions.** First, we provide a registered strategic-interaction example in which a fixed persona panel passes coarse condition-level and variance checks while producing aggregate continuation-probability point differences of +0.083 and +0.078; exact dependence-aware intervals show that these estimates remain compatible with materially larger positive or negative effects and do not establish equivalence (§4.1). Second, we identify and formalize the associated composition problem: finite-opportunity plug-in point estimates place 85%–96% of observed variation between prompts, while a latent-propensity sensitivity yields posterior median shares of 63%–71%; the binary boundary census is interval-method-sensitive, aggregate moments do not identify microstructure, and representation experiments show how wording and semantic labels govern the induced policies (§4.1–4.3). Third, we demonstrate an auditable reliability protocol—and its limit—through prospectively registered confirmatory claims, external chronology, exact replay, mechanical adjudication, and a public post-adjudication correction in which outside review overturns the favored persona-level interpretation without altering the frozen record (§4.4–4.5).

## 2. Related work

**Occupied territory, and where we sit.** Li and Ji [2026] establish across three model families, eleven interventions, and 59,508 participants that descriptive fit and intervention-effect accuracy can diverge, that prompt refinements improving realism do not reliably improve effect accuracy, and that errors vary with intervention logic, outcome structure, and attitude–behavior coupling. Persson, Schultzberg, and Ankargren [2026] formalize when LLM outcomes can serve as causal surrogates and why novel interventions still require human evidence. Lin et al. [2026] show that interventions can change the implicit simulated population even when explicit personas are fixed. Statistical-realism, persona-collapse, and state-versus-trait work further show that persona-conditioned populations can compress or misallocate heterogeneity [Xie et al. 2026; Harry et al. 2026; Xiao et al. 2026]. Our differentiation is therefore not the broad divergence or the existence of mechanisms. It is a registered decomposition of one common lightweight construction in strategic interaction: the same fixed explicit prompt panel is evaluated for marginal fit, finite-opportunity-corrected between/within composition, representation sensitivity, and response to a represented continuation-probability treatment, with exact prompt provenance and public inferential correction. The observed composition pattern is complementary to, not exclusive of, latent-user drift.

**LLM strategic behavior.** Akata et al. [2025] characterize repeated-game play modulated by prompts; Pal et al. [2026] elicit strategies from five models while varying continuation probability, payoffs, horizon knowledge, and framing; counterfactual-reasoning evaluations alter labels and payoff structures [Georgousis et al. 2026]; and “strategic robustness” has been defined as payoff-preserving invariance across narratives [Mousavi Davoudi et al. 2026]. These works establish that neither repeated games nor prompt/payoff perturbations are new. Our distinct combination is the fixed persona panel, the explicit between/within/response decomposition, exact prompt provenance, prospective registration of confirmatory claims, and mechanical adjudication followed by public inferential correction.

**Strong positive evidence, and a different estimand.** Ashokkumar, Hewitt, Ghezae, and Willer [2026] use study descriptions to forecast 469 effects from 70 preregistered, nationally representative survey experiments and find strong correlations with realized effects, alongside systematic effect-size overestimation and weaker performance in a megastudy archive. That is important contrary evidence against any blanket pessimism about LLMs in experimental science. It is also a forecasting task over studies rather than subject-level simulation of a response surface. Strong effect forecasting is compatible with the fixed-panel composition failure studied here and reinforces the prescription to validate a simulator on the exact response object for which it will be used.

**Synthetic participants and personas.** Bisbee et al. [2024] find plausible survey averages alongside compressed variance, distorted coefficients, and temporal drift; Boelaert et al. [2025] report excess homogeneity; Anthis et al. [2025] catalog diversity and generalization challenges; Hullman et al. [2026] propose statistical calibration for confirmatory use; and Park et al. [2024] show that rich interview conditioning can substantially outperform lightweight demographic/persona descriptions. Format sensitivity [Sclar et al. 2024], role-play framing [Shanahan et al. 2023], persona collapse [Xiao et al. 2026], state blindness [Harry et al. 2026], and reviews of persona-experiment transparency [Batzner et al. 2025] all caution against treating a persona string as a stable human analogue. RLHF-related diversity reduction is a possible mechanism for concentrated policies, not a mechanism identified by this design. Full map: docs/analysis/literature-map.md; differentiation table: docs/analysis/novelty-relationships.md.

### 3. Instrument and inferential units

The primary deployment is gpt-4.1 at temperature 0.7 with 16-token outputs and a fixed minimal behavioral-subject prompt containing no game-theory vocabulary or reasoning scaffold. Phase 5 prepends one sealed persona sentence to byte-identical task text, giving every persona cell an exact bare twin. The cross-vendor Gemini tier is descriptive; the original Claude Haiku candidate failed a registered entry gate and was replaced under an archived amendment. Environment randomness is seeded; provider-side generation is not claimed to be seeded. Every request, rendered prompt, completion, decoding configuration, round, and provenance record is archived.

#### 3.1 Sequential architecture and registration status

| Stage   | Primary question and role                                                                                                                       | Unit used in the paper                                          | Registration status                                                                                                                                                                                                               |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 1 | Initial prototype and naive behavioral claims; establishes the historical baseline, not current confirmatory evidence                           | provider calls / recorded decisions                             | post hoc instrument development                                                                                                                                                                                                   |
| Phase 2 | Mechanical re-adjudication and enforcement repair after the initial harness exposed analyst discretion                                          | archived runs and claim predicates                              | corrective, not prospective confirmation                                                                                                                                                                                          |
| Phase 3 | Bare GPT-4.1 configuration in repeated PD, framing, and RPS                                                                                     | episode, with historical seat-level summaries disclosed         | claims registered before Phase 3 data                                                                                                                                                                                             |
| Phase 4 | Representation robustness, X1/X2 wording extensions, counterfactual payoffs/labels, continuation-probability assays, adversaries, and sentinels | complete episode for current sensitivities                      | X1 was a sequentially registered, result-informed extension: Phase 3 motivated the test, while its prompts, sample size, and predicate were sealed before any X1 data; the remaining blocks were registered before their own data |
| Phase 5 | Sixteen sealed persona-prefix configurations crossed with the Phase 4 instruments; descriptive Gemini tier                                      | complete persona prompt for the fixed panel; episode beneath it | confirmatory predicates registered before Phase 5 data; post-adjudication sensitivities are explicitly unregistered                                                                                                               |

The present paper’s main empirical decomposition is Phase 5, interpreted using representation results from Phases 3–4. Phases 1–2 document instrument evolution and are not counted as prospective confirmation.

The full event store contains 5,505 completed runs, 54,276 round events, 108,552 seat-round decisions, and 36,251 archived provider-request events. The public Phase 4–5 replay contract covers 4,576 completed runs. A separate transactional ledger records 30,530 Phase 4–5 calls, 13,141,675 input tokens, and 45,247 output tokens; it excludes earlier phases and therefore must not be conflated with the full event-store request count.

Counts and definitions are reconciled in docs/analysis/submission/count-reconciliation.md.

Confirmatory claims were registered before the data that adjudicated them and were mechanically evaluated in a fixed vocabulary. The historical two-sided interiority rule used Clopper–Pearson bounds on seat-level round-one trials. Because two seats share an episode, the submission analysis additionally treats the complete episode as the independence unit. For an episode outcome  $Y \in \{0, 0.5, 1\}$ , the conservative exact sensitivity writes

$$Y = \frac{1}{2}\{1(Y \geq 0.5) + 1(Y = 1)\},$$

constructs simultaneous Clopper–Pearson intervals for the two episode-level binary components, and projects them onto  $E[Y]$ . This construction is conservative, does not assume seat independence, and does not collapse to zero uncertainty when all observed episodes agree. A Dirichlet–Jeffreys sensitivity uses the symmetric Dirichlet(0.5, 0.5, 0.5) prior on the probabilities of episode outcomes  $\{0, 0.5, 1\}$ ; posterior draws project  $E[Y] = 0.5q_{0.5} + q_1$ . The percentile cluster bootstrap is also retained as a post-adjudication sensitivity. The exact Clopper–Pearson projection is the conservative reference because it provides finite-sample coverage for the discrete episode mean; at  $n=6$  the percentile bootstrap has no comparable coverage guarantee and can understate uncertainty. Its degeneracy at exact corners is a symptom of that limitation, not by itself a false-positive mechanism for this strict interiority gate.

The hierarchy is deployment  $\rightarrow$  explicit persona prompt  $\rightarrow$  condition  $\rightarrow$  episode  $\rightarrow$  seat  $\rightarrow$  round  $\rightarrow$  provider request. Phase 5’s confirmatory unit is the complete persona sentence; name, age, occupation, and traits are bundled semantic treatments. Registered claims attach to the conditional finite-panel estimand for these sixteen prompts. Claims about a wider persona generator are exploratory at  $n = 16$ . Pairing the same explicit prompt across conditions identifies a prompt-indexed contrast, not necessarily a stable latent person’s treatment effect.

**Protocol glossary.** S2-absent and S2-present are the two registered repeated-game wording families: the latter includes the switch-bearing continuation sentence localized in X2, while the former omits that sentence. P5-1a is the registered fraction of selected persona–condition cells classified interior, with support for the corner-mixture predicate when that fraction is below 0.10; P5-1b is the registered between-persona dispersion comparison. P5-2 pools registered conflict cells and classifies whether choices follow task text or persona-conditioned direction. P5-3(a)—called clause (a) below—asks whether any persona  $\times$  wording pair has both continuation-probability cells interior and a positive slope lower bound; P5-3(b)—clause (b)—asks whether each persona lane rejects the bare configuration’s dominated, semantically attracted swap-cell option at a registered minimum rate. Historical alphanumeric verdicts remain visible even where post-adjudication sensitivities change their scientific interpretation.

**Phase 5 condition matrix.** The 96 Tier-A persona–condition units are the full cross of sixteen prompts with six conditions:

| Code        | Condition                              | Role in the paper                           |
|-------------|----------------------------------------|---------------------------------------------|
| rep-d10-s2a | repeated PD, $\delta=.10$ , S2 absent  | repeated-game level, variance, and response |
| rep-d10-s2p | repeated PD, $\delta=.10$ , S2 present | repeated-game level, variance, and response |
| rep-d90-s2a | repeated PD, $\delta=.90$ , S2 absent  | repeated-game level, variance, and response |

| Code         | Condition                              | Role in the paper                           |
|--------------|----------------------------------------|---------------------------------------------|
| rep-d90-s2p  | repeated PD, $\delta=.90$ , S2 present | repeated-game level, variance, and response |
| os-swap      | one-shot canonical-payoff label swap   | semantic-label/payoff conflict              |
| os-community | one-shot Community framing             | near-interior framing anchor                |

The registered P5-1a concept restricted its primary denominator to persona cells whose **exact recorded bare twin** failed the same interiority gate. An outcome-blind exact-twin completion fixed that set as rep-d90-s2a and os-swap: sixteen personas in each condition, hence 32 units. The exact Community twin passed the bare gate; the other three repeated-game cells lacked exact bare twins and entered only the unrestricted 96-cell secondary.

The machinery's boundary is explicit:

**The pipeline can enforce a registered predicate exactly; it cannot guarantee that the predicate represents a valid estimand, test family, or construct.**

## 4. Results

### 4.1 Coarse marginal checks pass while represented-treatment estimates remain imprecise

The preregistered leaning rule—at least two of agreeable, patient, and risk-averse—separates round-one cooperation by 0.5–0.7 in every non-swap cell. Because names, ages, occupations, and all trait descriptors vary in the complete sentence, this is a property of the registered prompts rather than a causal trait estimate. The preregistered P3-A3 broad-reference cooperation band is [0.36, 0.63]. The four repeated-game pool means are 0.349 (rep-d10-s2a), 0.427 (rep-d10-s2p), 0.432 (rep-d90-s2a), and 0.505 (rep-d90-s2p); only the S2-absent  $\delta=.10$  cell falls outside the band, by 0.011 below its lower boundary.

Raw cross-persona standard deviations range from 0.4241 to 0.4800. Correcting the plug-in estimates for finite episode counts leaves between-prompt SD point estimates of 0.4182, 0.4784, 0.4408, and 0.4323 in the cell order above. P5-1b used protocol-nonmatched human references mechanically implied from Dal Bó and Fréchette's [2011] R=40 Table 7 strategy-frequency estimates: SD=0.4122 for their  $\delta=.50$  panel and SD=0.3116 for their  $\delta=.75$  panel. The frozen ratio  $\rho=.75$  therefore gives thresholds 0.3092 for this study's  $\delta=.10$  cells and 0.2337 for its  $\delta=.90$  cells. All four plug-in estimates exceed those historical thresholds.

Three post-adjudication uncertainty views answer different questions. First, a **conditional fixed-panel episode bootstrap** retains the sixteen prompts and resamples their empirical episode distributions; its corrected-SD 95% intervals are [0.4122, 0.4391], [0.4696, 0.4916], [0.4279, 0.4654], and [0.4269, 0.4496]. Because an empirically unanimous six-episode cell resamples as a point mass, these intervals condition on the recorded boundary concentration and do not propagate uncertainty about latent prompt policies. Second, a **fixed-panel latent-propensity sensitivity** assigns each prompt/cell outcome distribution on {0, .5, 1} an independent Dirichlet(0.5, 0.5, 0.5) posterior. Its SD medians (95% intervals) are 0.3543 [0.3090, 0.3886], 0.3959 [0.3521, 0.4308], 0.3698 [0.3253, 0.4060], and 0.3620 [0.3179, 0.3970]; three of four lower bounds exceed the historical threshold, with rep-d10-s2a essentially on it (0.3090 versus 0.3092). Third, the **two-stage prompt+episode bootstrap**, which changes the estimand toward a hypothetical persona generator, yields corrected-SD intervals rep-d10-s2a [0.2724, 0.4879], rep-d10-s2p [0.3696, 0.5123], rep-d90-s2a [0.3457, 0.4890], and rep-d90-s2p [0.3345, 0.4847]. None of these post hoc sensitivities changes the frozen mechanical P5-1b verdict.

The corresponding plug-in between-prompt shares are 85.5%, 96.1%, 88.8%, and 90.2%. Conditional episode-bootstrap intervals are [82.0%, 93.8%], [94.6%, 98.9%], [86.7%, 94.6%], and [87.9%, 95.5%]. The latent-propensity posterior is materially wider and lower: medians 63.1%, 70.5%, 66.1%, and 66.5%, with 95% intervals [49.4%, 74.5%], [57.3%, 81.3%], [52.8%, 77.0%], and [52.8%, 77.7%]. Thus the robust statement is that between-prompt composition is substantial and likely dominant in this fixed panel; 85%–96% are plug-in point estimates, not fully uncertainty-adjusted population facts.

Across the represented continuation-probability treatment, the observed fixed-panel point differences are +0.083 for S2-absent wording and +0.078 for S2-present wording. Conservative exact simultaneous 95% intervals are [−0.171, +0.330] and [−0.181, +0.330] (rounded to three decimals). The point estimates are small on the unit scale, but the intervals permit effects as large as +0.33 and moderately negative effects; the experiment does not establish equivalence, a zero response, or a narrow upper bound. The treatment changes both the environment’s continuation process and the text used to communicate that process. Round-one choices therefore identify response to **continuation probability under a specified representation**, not a semantically neutral economic parameter: incentive and framing channels remain undecomposed.

These numbers have two distinct readings. For the finite archived panel, the contrasts +0.083 and +0.078 are exact descriptive arithmetic and require no sampling interval. For inference about the latent cooperation propensities of these sixteen configurations under repeated sampling, the conservative exact intervals remain wide. A plug-in or asymptotic clustered interval that treats zero observed within-cell variation as zero latent variance can be much narrower by construction; six agreeing episodes do not establish a deterministic boundary policy. A design-effect heuristic using six episodes per prompt and the plug-in between-share range 0.855–0.961 yields effective sample sizes of roughly 16.5–18.2 episode equivalents per condition, close to the sixteen prompt units. This is a planning heuristic, not the degrees of freedom of the exact procedure, but it identifies prompt count as the operative precision constraint.

Dal Bó and Fréchette [2011] remain useful only as protocol-nonmatched context: their treatments use different continuation probabilities and payoffs, monetary incentives, between-session assignment, and repeated supergames through which behavior changes with experience. Their pooled experienced contrast is substantially larger, while first-supergame ordering reverses. We make no matched magnitude or human-equivalence claim.

The registered Gemini tier was descriptive only and is excluded from these estimates because its endpoint showed documented non-stationarity. Across eight personas and three cells, recorded means ranged from 0 to 0.90 and 9/24 cells (37.5%) met the historical interiority rule, compared with 14/96 (14.6%) in the primary GPT-4.1 panel; several representation-channel effects also reversed direction across vendors. Because the endpoint was non-stationary and the evaluated panels differed, this is not a formal replication comparison. It is contrary descriptive evidence that the composition pattern is deployment-specific rather than universal. Complete values are in docs/analysis/figure-sources/p5-tierC-gemini.csv and the stability record.

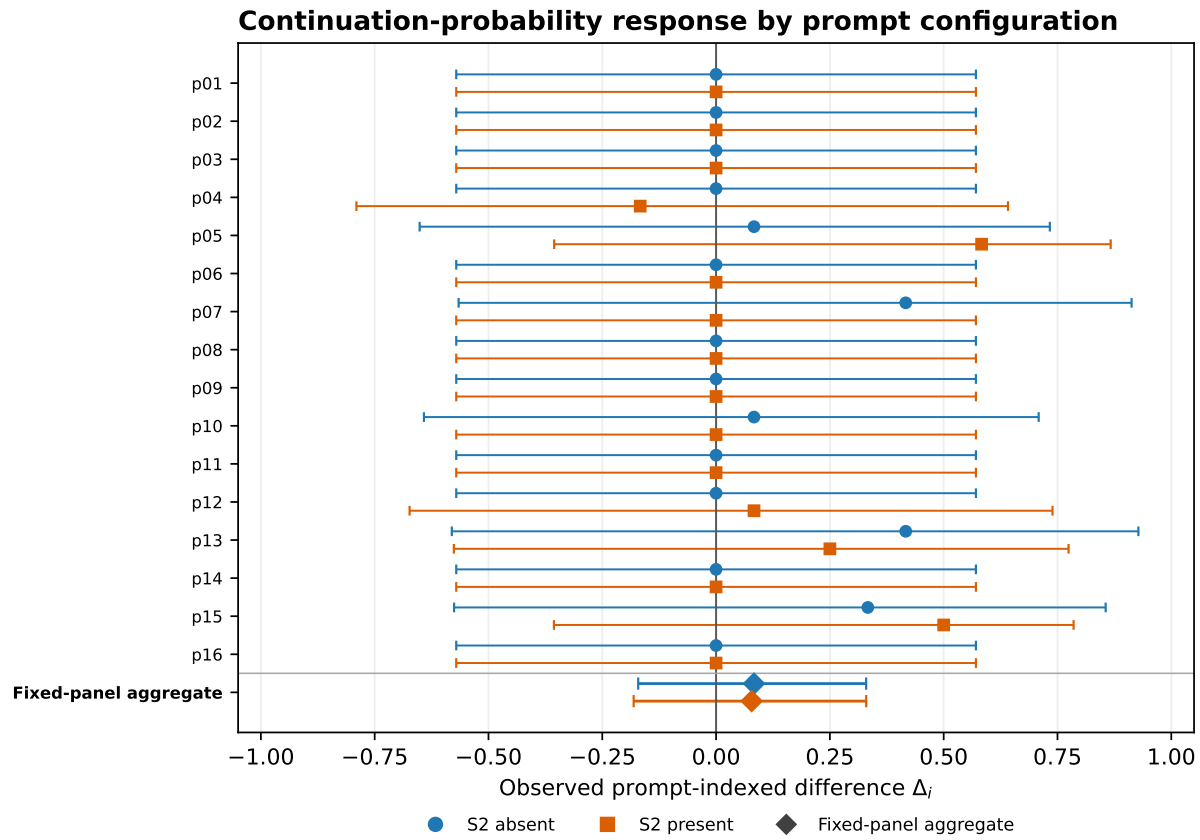


Figure 1. Prompt-indexed differences in round-one cooperation,  $\Delta_i = \hat{p}_i(\delta = .90) - \hat{p}_i(\delta = .10)$ , for both registered wording families. The series are vertically dodged so coincident values and intervals remain separately visible. Bars are conservative exact simultaneous 95% intervals with complete episodes as the unit; observed corners retain non-zero uncertainty. Diamonds on the **Fixed-panel aggregate** row show the two wording-family estimates in their corresponding colors. Rows at  $\Delta_i = 0$  can reflect boundary concentration in both recorded cells; they are not precise evidence of homogeneous response or no effect.

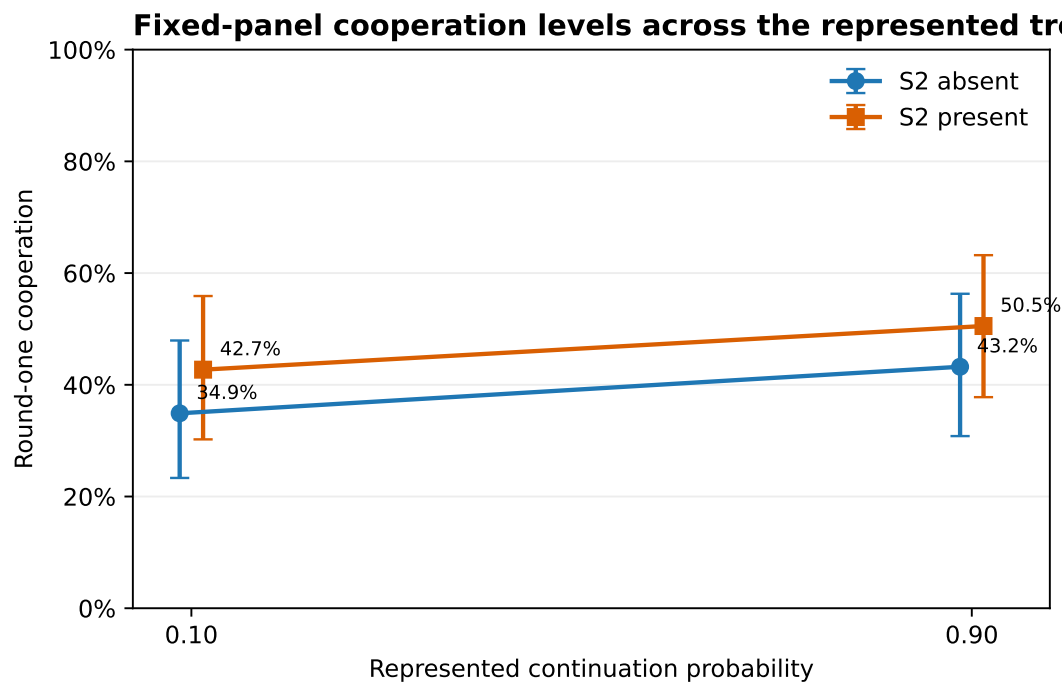


Figure 2. Fixed-panel round-one cooperation by represented continuation-probability condition and wording family. Series are horizontally dodged; point labels and the y-axis are percentages. Error bars are conservative exact condition intervals. Lines connect conditions for orientation only and do not imply a precise, semantically isolated incentive effect.

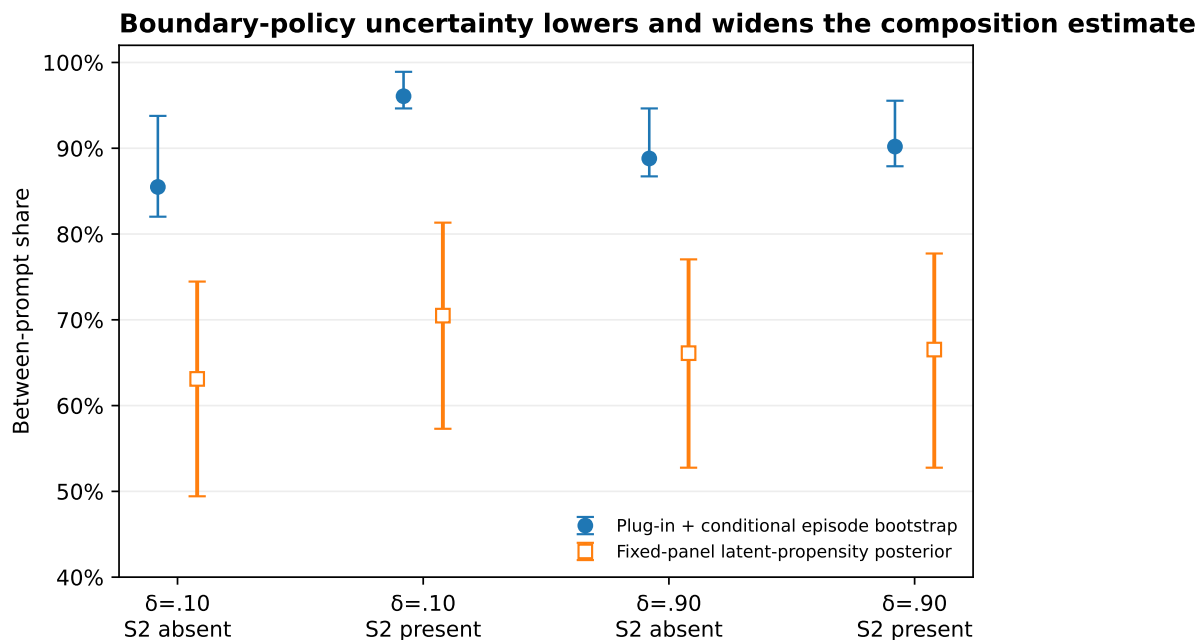


Figure 3. Between-prompt share of episode-level variation in each repeated-game cell. Circles and thin intervals show the finite-opportunity-corrected plug-in estimates with conditional episode-bootstrap intervals. Offset squares and thicker intervals show the fixed-panel Dirichlet–Jeffreys latent-propensity posterior medians and 95% intervals, which propagate uncertainty from empirically unanimous cells. The wider two-stage prompt+episode bootstrap changes the estimand toward a persona generator and is reported in text.

Across all six Phase 5 conditions, the historical seat-level rule classifies 14/96 persona-condition cells interior, the conservative exact episode-level interval 11/96, and the Dirichlet-Jeffreys sensitivity 19/96. In the registered 32-unit restricted set (rep-d90-s2a plus os-swap), the corresponding counts are 3/32, 2/32, and 5/32. The historical 3/32 fraction falls just below the frozen 0.10 threshold; the Bayesian sensitivity would not. Thus the binary verdict is not invariant to interval construction. The continuous uncertainty analyses above provide the less brittle description.

#### 4.2 What the marginal checks cannot identify

Let  $p_i(d) = E[Y \mid i, d]$ , where  $i$  indexes the complete explicit persona prompt and  $d$  the experimental condition.

**Proposition A: broad bands only partially identify the aggregate contrast.** If a synthetic condition mean is accepted within tolerance  $\epsilon_d$  of a reference mean in each condition, then

$$|\Delta^S - \Delta^H| \leq \epsilon_0 + \epsilon_1.$$

Equivalently, accepted bands  $[\ell_0, u_0]$  and  $[\ell_1, u_1]$  imply only

$$\Delta^S \in [\ell_1 - u_0, u_1 - \ell_0].$$

Exact condition-specific mean matching would force the aggregate effect by the identity  $\Delta = \mu_1 - \mu_0$ ; the empirical failure lives in the slack of coarse criteria.

**Proposition B: aggregate moments do not identify microstructure or response coupling.** This is an application of the law of total variance and classical Fréchet-Hoeffding/Sklar coupling results [Hoeffding 1940; Sklar 1959] to synthetic-participant validation, not a new probability theorem. Mean and total variance do not identify how variation is divided between prompt configurations and repeated draws, nor do they identify distributional shape or boundary concentration. Even exact condition-specific distributions do not identify the cross-condition coupling and therefore do not identify the distribution of prompt-indexed responses  $\Delta_i = p_i(1) - p_i(0)$ . Reusing an explicit persona string supplies one prompt-indexed coupling, but interpreting it as a stable synthetic individual’s potential-outcome contrast requires latent-person invariance, which this study does not test.

The study therefore identifies a composition pattern in one fixed prompt panel. It does not establish that humans have a different microstructure, that RLHF caused the pattern, that latent-user drift is absent, or that the same pattern occurs across persona generators.

#### 4.3 Control-channel interactions

For the bare configuration, round-one cooperation was 0.000 at every registered continuation probability. X2 mechanically decomposed the v1 and v2a prompt bundles into six rendered sentence/block spans and constructed forward and reverse ladders by replacing one complete span at a time. The selected S2 operation replaced and repositioned the v1 sentence “After every round there is a {deltaPct}% chance the session continues with another round” with the v2a first-line sentence “At the end of each round there is a {deltaPct}% chance that the session goes on for one more round.” Here **switch-bearing** means the span whose adjacent substitution produced the largest preregistered ladder gap and then passed held-out confirmation; it is not

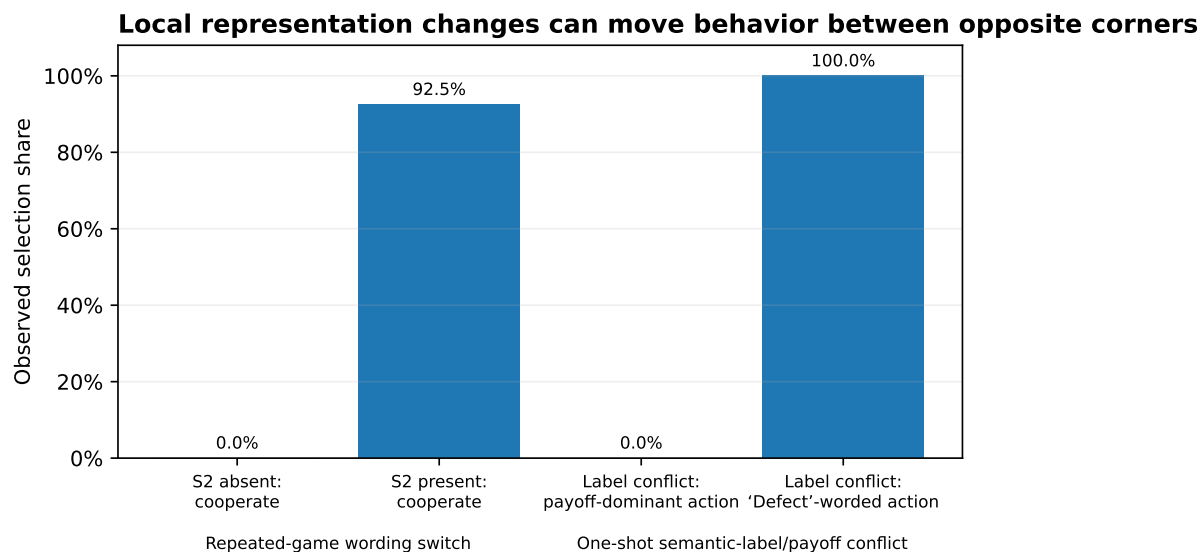
a strategy-switch instruction. Screening used ten episodes per rung. The exact S2 minimal pair was then tested at temperature 0.7 on 20 fresh episodes per side (seeds 2953–2972, disjoint from screening), moving held-out cooperation from 0/40 to 37/40. Because S2 combined wording and position as one atomic operation, the design does not separate those components or eliminate every positional interaction. The same wording factor was null in one-shot play, showing that text effects depend on strategic context.

In the label-swap conflict cell, canonical payoffs were held fixed while the displayed words “Cooperate” and “Defect” were attached to opposite strategic roles. The bare configuration selected the cooperation-worded option 0/40 times, choosing the strictly dominated role whenever it carried the word “Defect,” while responding strongly to payoff changes when semantic labels did not oppose them. The precise conclusion is conditional: semantic labels can override payoff dominance in a registered conflict cell; payoff sensitivity is representation-dependent, not absent.

Persona conditioning produces two observed contrasts, but they are not fully factorially separable. Differences among complete persona prompts produce the 0.5–0.7 leaning gaps. Adding any tested persona string reverses the bare swap-cell choice: all sixteen personas overwhelmingly select the cooperation-worded/payoff-dominant option. Because no non-semantic prefix matched for length, punctuation, and position was run, this prefix contrast cannot isolate semantic persona content from generic sequence-length or displacement effects. The choice result is also mechanism-confounded because word and payoff point to the same action.

P5-3(b)’s 24 evaluable persona  $\times$  temperature lanes comprise all sixteen personas at  $T=0.7$  plus the registered sweep subset p02, p06, p11, and p15 at each of  $T=1.0$  and  $T=1.3$  ( $16+4+4=24$ ). Under simultaneous episode-exact familywise bounds, every lane retains a lower bound above the frozen 0.20 threshold; the minimum is 0.462. This establishes the choice pattern, not whether incentives or lexical attraction caused it.

The pooled P5-2 task-consistent share is 90 of 704 seat decisions across 352 independent episodes. Equivalently, the pooled episode-mean total is  $45/352=0.128$ , with conservative exact episode-level 95% interval  $[0.092, 0.172]$ , retaining the historical persona-dominant classification. Every repeated-game conflict subcell is mixed under the exact episode interval. Only the swap cell is individually persona-dominant, with task-consistent share 0 and interval  $[0, 0.027]$ . The pooled verdict is therefore carried entirely by the word/payoff-confounded cell. We describe these findings as **control-channel interactions**, not a fixed hierarchy.



*Figure 4. Two distinct representation interventions in the bare configuration. In repeated play, the registered S2 wording-and-position operation moved observed cooperation from 0/40 to 37/40. In the one-shot label conflict, the payoff-dominant action was never chosen when the dominated action carried the displayed word “Defect.” The bars report selection shares, not a common causal estimand.*

#### 4.4 The favored persona-level result is not prospectively confirmed; the archived family is underpowered

Under the historical seat-level rule, persona p13 moved from 0.333 cooperation at  $\delta=.10$  to 0.750 at  $\delta=.90$  and passed a per-candidate lower-bound test. The rule searched multiple persona  $\times$  wording candidates and fired on any pass without declared family-level error control. External review identified that defect.

Three 200,000-permutation gate constructions are now reported; all permutation p-values use the add-one convention,  $\hat{p} = (r + 1)/(B + 1)$ , with exact Monte Carlo intervals. Under the historical seat-level gate, p13 remains the maximum at +0.4167, with familywise  $p = 0.059230$ , Monte Carlo 95% interval [0.058194, 0.060268]. Under the percentile episode-cluster-bootstrap sensitivity, p13 also remains the maximum and  $p = 0.043455$ , interval [0.042561, 0.044353]. The exact projection is the conservative reference because it has finite-sample coverage for the discrete episode mean; the percentile bootstrap is retained symmetrically but has no comparable small-sample coverage guarantee. Under the conservative exact-episode gate, p13 is ineligible: its low- $\delta$  lower bound falls below 0.05 and its high- $\delta$  upper bound exceeds 0.95. Only p04/s2p and p05/s2a pass both gates; the largest eligible slope belongs to p05/s2a (+0.0833), with familywise  $p = 0.773206$ , interval [0.771363, 0.775039].

For p13/s2a, the percentile bootstrap admitted both conditions as interior— $\delta=.10$ : [0.083, 0.667];  $\delta=.90$ : [0.583, 0.917]—whereas the conservative exact projection rejected both—[0.047, 0.800] and [0.287, 0.954], respectively. Neither recorded cell was at an exact corner. The eligibility difference therefore arises from small-sample interval width and coverage behavior, not from a corner interval falsely passing the gate.

The complete data-dependent gate is dynamically reapplied within every permutation, not frozen from the observed-data mask. The implementation precomputes 56 possible-composition gate values and performs 25,600,000 condition-gate lookup applications at  $B=200,000$ . In a deliberately incorrect comparison that froze the observed-data mask, the maximum statistic differed from the dynamic procedure in 718 of 5,000 null draws (14.4%), showing that reapplication materially changes the reference distribution. Lookup/direct parity and the regression are recorded in docs/analysis/submission/round5/round5-review-audit.md.

An exhaustive attainability audit shows that, with six episodes per condition, 12 of the 28 possible episode-value compositions pass the exact gate. Their sample means range from 0.333 to 0.667, but eligibility depends on the full 0, 0.5, 1 composition rather than on the mean alone. Two eligible cells can therefore differ by at most 0.333. Under the archived 32-candidate null structure, the add-one familywise tail probability at that maximum attainable slope is 0.075040, with Monte Carlo 95% interval [0.073884, 0.076198]; no exact-gate result in this archived family can reach 0.05. The exact analysis is therefore a valid dependence-aware sensitivity but not a powered disconfirmation of a p13-sized capability claim. The record neither prospectively confirms nor decisively disconfirms p13; it identifies a replication target whose next test must be sized prospectively (Appendix A.4).

These variants were specified and executed together only after external review exposed the original defects. No familywise gate was registered at the original freeze, and the repository contains no seal-before-compute record for these sensitivities. The analyses ran in GitHub Actions against archived databases with fixed seeds, and their outputs were committed regardless of direction. Accordingly, the favorable  $p = 0.043455$  variant cannot create prospective confirmation, just as the other variants cannot retroactively strengthen the

frozen claim. The historical mechanical P5-3 verdict remains visible because clause (b) fired strongly and p13 passed the rule as frozen. Scientifically, p13 is a replication target, not a finding. Clause (a) did not prospectively establish the capability claim because the frozen search lacked family control; the conservative post-adjudication procedure is too underpowered at  $n=6$  to provide decisive evidence against it.

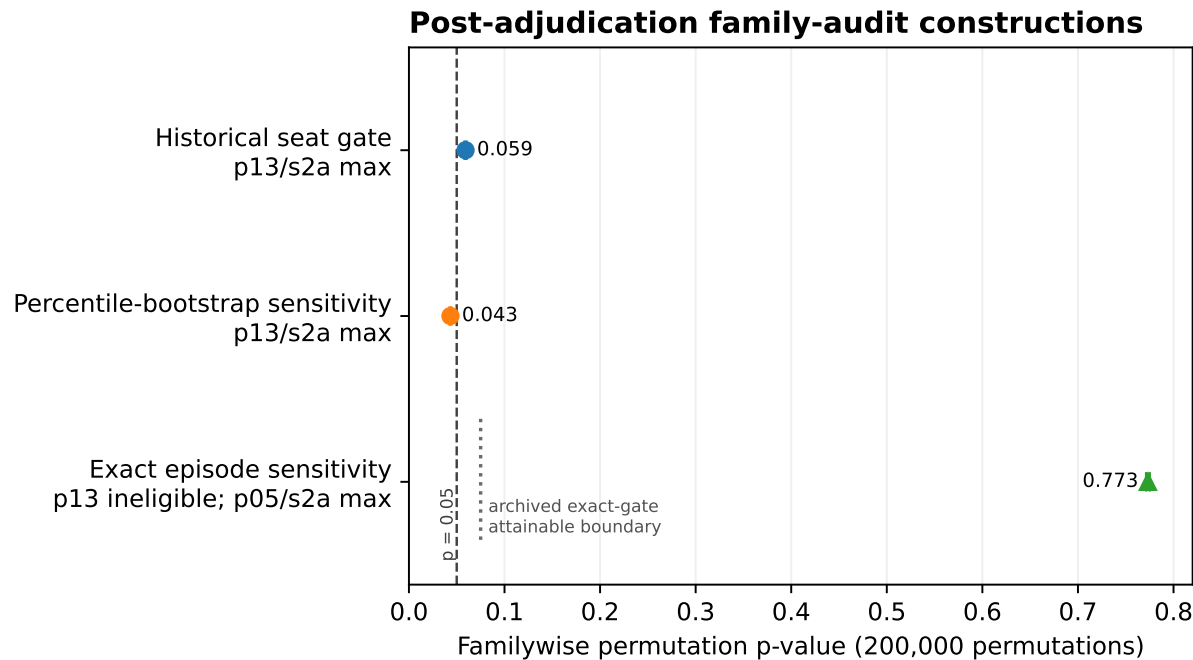


Figure 5. Post-adjudication familywise constructions. The first two points are p13/s2a under the historical and percentile-bootstrap gates. Under the conservative exact-episode gate, p13 is ineligible; the third point is the familywise result for the largest eligible candidate, p05/s2a. The dotted line at  $p = 0.075040$  marks the estimated minimum attainable familywise  $p$ -value for the archived  $n = 6$ , 32-candidate exact-gate design and applies only to that construction. None of the procedures was registered at the original freeze; no point creates prospective confirmation.

#### 4.5 Auditability

Twelve registered author predictions were refuted by data and published. Four underspecified analysis choices were resolved outcome-blind. A two-sided gate blocked a false ceiling conclusion. Sentinels detected a time-indexed behavioral discontinuity in an unversioned endpoint and exposed a monitoring gap that was repaired with an attestation gate. The public capsule replays all 4,576 Phase 4–5 runs exactly with zero live model calls. External review then found the family-error and dependence defects discussed above; the archived record made both diagnosable and correctable.

The claim is bounded: the machinery is procedurally exact and extensively auditable. Its strongest achievement is not self-validation but preservation of enough provenance for outsiders to identify where procedural correctness stopped short of statistical validity.

5. Discussion

5.1 Implications

For synthetic-participant practice, broad marginal resemblance is a weak validation target. Lightweight conditioning can generate plausible aggregate levels and substantial cross-prompt dispersion while yielding small observed treatment-response point estimates whose uncertainty remains wide. Validation should therefore report response surfaces over declared interventions and representation families, together with assay-sensitivity checks, dependence-aware uncertainty, model/provider provenance, and temporal monitoring. Statistical calibration [Hullman et al. 2026], causal-surrogacy assumptions [Persson et al. 2026], and latent-drift diagnostics [Lin et al. 2026] are complementary rather than competing safeguards.

The result can be read as a synthetic-subject analogue of the reduced-form/structural distinction associated with the Lucas critique [Lucas 1976]: fitting or selecting for aggregate resemblance need not identify behavior under a changed treatment. In psychometric terms, it is a construct-validity and assay-sensitivity problem [Cronbach & Meehl 1955; ICH E10; Temple & Ellenberg 2000]. Agent-based modeling’s equifinality and pattern-oriented validation provide a related analogy [Windrum et al. 2007; Grimm et al. 2005]. These are organizing analogies, not claims that the study literally estimates a structural economic model or validates a human measurement model.

The results also show why binary certification language should be used sparingly. The historical P5-1a predicate passes under its frozen seat-level rule and under the conservative episode-exact interval, but fails under a reasonable Dirichlet–Jeffreys sensitivity. The underlying continuous evidence is clearer than the thresholded label: plug-in estimates assign most observed variation between prompts, and the fixed-panel latent-propensity sensitivity still places posterior medians above one-half while substantially widening and lowering the intervals.

For deployment, behavior that can be rewritten by a sentence, an action token, or an identity prefix is safety-relevant. The results do not imply that language always dominates incentives: in some cells numerical payoffs move behavior strongly, and precedence depends on representation and conflict structure.

5.2 The precommitted discussion, and what changed after review

The Phase 5 discussion was sealed before its data existed. Excerpt (full text in supplement; sha1f1d7de9...e356):

“The headline of Phase 5 is an existence result the program registered against itself: at least one persona in the sealed sixteen passed the two-sided assay gate and showed the registered signature of incentive sensitivity. The author’s registered prediction—that none would—is refuted, and the refutation is the finding. ... The capability was recoverable by content-side conditioning... The scope of the claim is deliberately narrow.”

| Precommitted interpretation                                                          | Current status after external review and zero-call reanalysis                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| At least one persona establishes an unconfounded incentive-response existence result | <b>Not prospectively established.</b> The frozen search lacked family control, and the post-adjudication procedures were method-dependent and unregistered. The conservative exact procedure cannot reach conventional familywise rejection in the archived n=6 design. See §4.4; p13 remains a replication target. |

|                                                                                         | Current status after external review and zero-call reanalysis                                                                                                                                                    |
|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Precommitted interpretation                                                             |                                                                                                                                                                                                                  |
| “No game-relevant instruction—trait words only”                                         | Restated precisely: no explicit game terminology, action recommendation, or payoff reference; traits are strategically relevant information, and name, age, and occupation are uncontrolled semantic treatments. |
| Persona framing “contested or beat” task switches                                       | Pooled choice result survives exact episode inference, but every repeated conflict subcell is mixed; the pooled dominance classification is entirely carried by the word/payoff-confounded swap cell.            |
| Bare corners characterize the configuration rather than the model’s capability envelope | Retained only as a future hypothesis. The p13 evidence no longer supports it; clause (b) demonstrates a robust choice reversal but does not identify incentive sensitivity.                                      |

The sealed text remains unchanged because its evidentiary value lies partly in making interpretive error visible. The correction is additive and explicit rather than silently rewritten.

## 6. Limitations

The primary evidence comes from one model deployment and sixteen complete persona prompts. Population-level generalization is weakly identified. The boundary census depends on interval construction: the historical seat rule, exact episode projection, and Dirichlet–Jeffreys sensitivity produce different counts, although the corrected variance analysis consistently locates most estimated variation between prompts rather than within them. The exact episode interval is conservative, while the Bayesian sensitivity depends on its prior. The variance correction is a fixed-panel method-of-moments/bootstrap sensitivity, not a matched estimate of human latent heterogeneity.

The two aggregate continuation-probability point estimates are small on the unit scale, but their conservative intervals extend to approximately +0.33 and moderately negative values; the data do not establish equivalence, incentive insensitivity, or a narrow maximum effect. Explicit persona strings are paired across conditions, but latent-person invariance is untested. Clause (b) remains word/payoff-confounded. Human references are published and protocol-nonmatched. The Dal Bó–Fréchette microdata pipeline is fixture-tested but the licensed package remains login-gated; Q5 is therefore deferred and nonblocking under the current internal claim. The temperature observation lacks an identified mechanism, the high-temperature  $\delta$  interaction is not estimable under the registered design, and the Gemini tier is descriptive under documented endpoint non-stationarity. Because choice entropy fell rather than rose over the registered temperature sweep, boundary concentration at  $T=0.7$  cannot be attributed to persona conditioning in isolation; a prospectively crossed temperature  $\times$  persona-prefix design is needed to partition those contributions. Because choice entropy fell rather than rose over the registered temperature sweep, boundary concentration at  $T=0.7$  cannot be attributed to persona conditioning in isolation; a prospectively crossed temperature  $\times$  persona-prefix design is needed to partition those contributions.

The prefix and continuation-treatment construct limits are described in §4.1 and §4.3: no format-matched neutral prefix was run, and the continuation process was manipulated together with its textual representation. The six-episode exact-gate family analysis is underpowered by construction, as quantified in §4.4, and should not be read as evidence that p13 has no response.

The post-adjudication family analyses were specified after reviewers found the original defects and were not sealed before computation. Fixed seeds, Actions execution, complete output retention, and public review

constrain discretion but do not convert the analyses into preregistered confirmation. The natural next test is a newly registered replication with episode-level gates and family control fixed before data.

## 7. Reproducibility

The repository is public. The capsule replays 4,576/4,576 Phase 4–5 runs byte-exact with zero credentials and zero live model calls. Prompt registries, freeze packets, claim ledgers, adjudication records, external timestamp proofs, post-adjudication sensitivity scripts, generated result tables, draft history, and reviewer records are versioned in the repository. The submission analyses run in GitHub Actions directly against archived databases and commit their outputs to the review branch; they make no provider calls and alter no sealed source artifact.

For Phase 4–5, the event record contains 30,421 normal `llm.requested` events and 30,397 `llm.responded` events; the 24-event difference is the disclosed set of provider-failure partials. The response records contain complete rendered prompts, bundle and request-body SHA-256 values, engine commit and provider route, raw completion text, and provider response IDs for all 30,397 responses. The separate budget ledger records 30,530 call attempts, including Phase 4 infrastructure and spend-accounting entries outside the standard request-event stream, so it is not row-identical to either event count. The live adapter asserted that its actual deterministic request-body hash matched the recorded mirror. Individual raw completion payloads were not separately hash-chained or provider-attested at receipt, and the full provider JSON object was not retained. Capsule checksum manifests and external timestamps make the later released database snapshot tamper-evident relative to that snapshot; replay does not prove that no alteration occurred between provider receipt and snapshot sealing.

The original `scope-seal.md` was sealed byte-for-byte with a header that still reads “PROPOSED—UNSEALED.” It is not edited because its hash is part of the Phase 5 seal. A living status addendum points to the sealing event, manifest hash, and stopping-rule evidence.

## 8. Attribution

The human author selected the research questions, approved the registered designs, adjudicated which reviewer recommendations to adopt, and accepts responsibility for every claim. The autonomous pipeline executed registration, dispatch, adjudication, and replay as apparatus. AI reviewers supplied adversarial analysis that materially changed the manuscript, including the family-error diagnosis, the condition-mean identity, the dependence-unit critique, and the latent-person-invariance correction.

The role of one round-2 reviewer subsequently expanded from critique to specification of the post-adjudication analyses and management of their integration. Those analyses were executed by GitHub Actions against the archived databases; resulting commits were authored by Yohei Nakajima or the Actions bot. A later reviewer independently checked the branch through an anonymous blobless clone against the generated JSON and commit history. Review prompts, role changes, adopted recommendations, and the independent verification memo are archived under `docs/reviews/`. Venue-specific AI-assistance language will be conformed at submission.

## Appendix A — Cutting room and correction ledger

### A.1 Findings retained outside the main arc

- **Entropy versus temperature:** choice entropy fell as decoding temperature rose, including on a matched-unit lattice. This is real in the recorded data but mechanism-free and remains supplementary.
- **RPS role-attached rock bias:** survived neutral symbols and randomized display order, with a cross-vendor sign reversal. Supplementary because it adds another game without strengthening the main claim.
- **Adversary-suite secondaries:** order-2 tracking exploited the primary configuration while a first-order tracker lost; the registered WSLs-targeter prediction failed. These support opponent contingency but require a separate technical appendix.
- **Drift case study:** retained as a full repository artifact and summarized only in §4.5.
- **Claude Haiku entry-gate failure:** preserved as evidence that eligibility to act as a behavioral subject is itself an empirical question.

### A.2 Framings set aside

- “Level-Matching Is Cheap” and “Moment Matching Is Cheap” were retired because exact condition-specific mean matching would recover the aggregate contrast by identity.
- “Corner Mixtures” was removed from the title because the binary corner census is interval-method-sensitive. The less brittle load-bearing construct is the persona-mixture composition and corrected between-prompt share.
- Discovery-forward “One Persona in Sixteen” was rejected because the frozen rule lacked prospective family control and the conservative post-adjudication procedure is too underpowered for decisive confirmation or disconfirmation.
- Metascience-first, safety-first, prompt-stack, and temperature-anomaly papers remain possible spin-outs.

### A.3 Analyses not run

The balanced-payoff × persona cell remains the highest-value de-confounding experiment. Other follow-ups include a prospectively family-controlled p13 replication, broader persona sampling, cross-model persona transfer, richer interview-conditioned panels, a matched human arm, and additional wording interpolation. No new arm was added after the scope seal.

The Dal Bó–Fréchette microdata pipeline is written and fixture-tested, but the official package remains login-gated. The statistical submission gate is complete with this Q5 contextualization deferred. Until licensed data are supplied, only published table values are used and every comparison remains labeled nonmatched.

### A.4 Prospective replication and open design choices

A Phase 6 replication should not inherit the full 32-candidate search by default. It should preselect the replication target or a small declared family, use complete episodes as the unit, declare family control before data, and size the design by simulation against the exact decision rule. At  $n=6$ , 12 of the 28 possible episode-value compositions pass the exact gate; their means span  $[0.333, 0.667]$ , but eligibility depends on the full  $\{0, 0.5, 1\}$  composition rather than the mean alone. The largest eligible slope is 0.333; under the archived family, even that maximum has add-one tail probability 0.075040 with Monte Carlo 95% interval  $[0.073884, 0.076198]$ . The present design is therefore incapable of conventional exact-family rejection, not merely imprecise. Prospective planning simulations in docs/analysis/submission/round5/ show how power changes with episode

count and family size, but the registered Phase 6 calculation must use its declared dependence model, effect margin, and exact decision rule.

The principal open design choices are the smallest scientifically relevant response, candidate-family size, format-matched prefix controls, a continuation-probability  $\times$  wording factorial, and whether the next study targets one registered configuration or a broader persona generator.

### A.5 Correction ledger

| Change                                                                                                                                                                                                                                                                                                                | Source                                                   | Category                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------|
| p13 confirmatory $\rightarrow$ replication target                                                                                                                                                                                                                                                                     | External review of family error                          | Inference                 |
| All three 200,000-permutation variants reported: historical seat gate $p = 0.059230$ ; percentile-bootstrap sensitivity $p = 0.043455$ ; conservative exact-episode sensitivity makes p13 ineligible and attributes the maximum eligible result to p05/s2a, $p = 0.773206$ ; none was prospectively family-controlled | Independent verification + zero-call submission analyses | Inference transparency    |
| Post-adjudication variants explicitly labeled unregistered at the original freeze; fixed-seed Actions execution and complete output retention documented                                                                                                                                                              | Independent verification                                 | Chronology and discretion |
| Seat-level historical gate supplemented by exact episode-level CP projection                                                                                                                                                                                                                                          | External dependence critique + zero-call analysis        | Statistical unit          |
| P5-1a historical 3/32; exact episode 2/32; Dirichlet sensitivity 5/32                                                                                                                                                                                                                                                 | Zero-call submission analysis                            | Robustness                |
| Percentile cluster bootstrap retained as a sensitivity; exact projection used as the conservative finite-sample coverage reference                                                                                                                                                                                    | Statistical audit                                        | Statistical validity      |
| Raw SDs corrected for finite episode counts; corrected SD 0.418–0.478 and between share 85%–96%                                                                                                                                                                                                                       | Zero-call variance analysis                              | Variance interpretation   |
| Prompt-indexed $\Delta_i$ figure added with exact non-degenerate intervals, aggregate intervals, and latent-coupling caveat                                                                                                                                                                                           | Independent review recommendation                        | Interpretability          |
| P5-2 pooled result survives, but exact episode analysis makes every repeated conflict subcell mixed; swap alone carries dominance                                                                                                                                                                                     | Zero-call submission analysis                            | Construct interpretation  |
| Clause (b) survives simultaneous familywise exact episode bounds in all 24 lanes; minimum lower bound 0.462                                                                                                                                                                                                           | Zero-call submission analysis                            | Statistical robustness    |

| Change                                                                                                                                                               | Source                               | Category                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|--------------------------|
| “ $\delta$ -matched” and N-fold human comparisons retired                                                                                                            | Comparator audit                     | Comparator validity      |
| Lin-style latent drift reframed as potentially coexisting, not excluded                                                                                              | Literature verification              | Mechanism scope          |
| Exact condition-level matching distinguished from broad-band checks                                                                                                  | Identification audit                 | Theory                   |
| Full-store and ledger counts reconciled: 36,251 request events versus 30,530 Phase 4–5 ledger calls; 5,505 archived completed runs versus 4,576 replay-contract runs | Zero-call count analysis             | Provenance               |
| “Text-determined, not payoff-determined” replaced by conditional semantic dominance                                                                                  | Reviewer critique                    | Precision                |
| Complete persona sentence treated as the unit; trait causality demoted                                                                                               | Construct audit                      | Scope                    |
| Sealed discussion preserved with correction table rather than silently edited                                                                                        | Transparency design                  | Research integrity       |
| Confirmatory registration language narrowed to claims registered before their adjudicating data                                                                      | Independent verification             | Registration accuracy    |
| “Weak observed response” replaced by small, imprecisely estimated point differences with intervals extending to about +0.33                                          | Round 4 review                       | Uncertainty discipline   |
| Li–Ji differentiation narrowed after re-reading its expanded mechanism analysis                                                                                      | Round 4 literature verification      | Novelty scope            |
| Ashokkumar–Hewitt et al. restored as the strongest positive contrary result and distinguished as study-level forecasting                                             | Round 4 review                       | Contrary evidence        |
| Aggregate and per-prompt response estimates added to the machine-readable submission summary                                                                         | Round 4 review                       | Reproducibility          |
| Dynamic gate reapplication documented and regression-tested; 25.6 million lookup applications at B=200,000                                                           | Explore Science B1 + zero-call audit | Familywise inference     |
| Exact-gate attainability and prospective-power audit added; p13 reframed as neither confirmed nor decisively disconfirmed                                            | Explore Science B3 + zero-call audit | Power and interpretation |
| Percentile-bootstrap rationale corrected from corner misclassification to lack of small-sample coverage guarantee                                                    | Explore Science B5                   | Statistical validity     |

| Change                                                                                                                                                                | Source                          | Category                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|--------------------------------|
| Persona-prefix effect qualified as format/content-confounded; $\delta$ reframed as a represented treatment                                                            | Explore Science B2/B7           | Construct validity             |
| Request/response provenance and archive tamper-evidence boundary audited and stated                                                                                   | Explore Science A2              | Provenance                     |
| Phase table, protocol glossary, span-ladder specification, Gemini pointer, and probability-theory grounding added                                                     | Explore Science A1/B4/B6/B8/B9  | Self-contained reporting       |
| Figures 1 and 5 corrected for aggregate markers and candidate attribution                                                                                             | Explore Science C1/C2           | Figure integrity               |
| Human SD references, thresholds, broad-reference band, failing cell, condition matrix, restricted denominator, P5-2 counts, and 24-lane construction reported in text | Explore Science v10 C1/C2/C4–C7 | Reporting completeness         |
| Fixed-panel Dirichlet–Jeffreys latent-propensity sensitivity added to propagate boundary-cell uncertainty without changing the prompt panel                           | Explore Science v10 C3          | Uncertainty interpretation     |
| Figures 1, 2, 3, and 5 redrawn for dodging, unit consistency, dual uncertainty views, and reference-line labels                                                       | Explore Science v10 D1–D4       | Figure integrity               |
| Bare-configuration and represented-treatment qualifiers added to abstract; S2 sentence quoted and defined                                                             | Explore Science v10 A2/A3/C10   | Construct precision            |
| Abstract registration scope restricted to confirmatory Phases 3–5; X1 labeled a sequentially registered result-informed extension                                     | Claude repository review        | Registration clarity           |
| Dynamic-versus-static gate divergence (718/5,000; 14.4%) and concrete p13 interval disagreement reported                                                              | Claude repository review        | Inferential transparency       |
| Title changed from incentive-response to treatment-response; Gemini framed as deployment-specific contrary evidence                                                   | Claude repository review        | Construct and scope discipline |
| Figure 1 aggregate row/legend, Figure 2 condition intervals, and Figure 5 archived-design attainability boundary added                                                | Claude repository review        | Figure integrity               |
| Review artifact identity metadata and venue-neutral appendix added                                                                                                    | Claude repository review        | Review provenance              |

| Change                                                                                                                                     | Source           | Category                     |
|--------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------|
| Archived-record contrasts separated from latent-propensity inference; “cheap” defined as validation without estimating the response object | Claude v8 review | Estimand and framing clarity |

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